ISI
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Test Code : MEI/MEII 2000

2000

Syllabus for MEI

Matrix Algebra : Matrices and Vectors, Matrix Operations, Determinants, Non-singular Inversion, Cramer's rule.

Calculus : Limits, Continuity, Differentiation of functions of one or more variables. Product rule. Partial and total derivatives. Derivatives of implicit functions. unconstrained optimisation (first and second order conditions for extrema of functions of single variable and several variables). Taylor series. Definite and indefinite integrals standard formulae, integration by parts and integration by substitution. Differential equations.

Constrained optimisation of functions of a single variable

Theory of sequences and series.

Linear programming : Formulation. Statements of primal and dual problems. Graphical solutions.

Theory of polynomial equations (up to third degree)

Sample Questions MEI (Mathematics) 2000

Each question carries 4 marks.

1. The domain of definition of the function $f(x) = \sqrt{x} + \sqrt{3-x} + \sqrt{x^2 - 4x}$ is

(A) $[0, 3]$. (B) $(0, 3)$. (C) $\{0\}$ (D) none of these.

2. The sum of the first n terms of the series

$\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$

is equal to

(A) $2^n - n - 1$. (B) $1 - 2^{-n}$ (C) $n + 2^{-n} - 1$ (D) none of these.

3. The matrix $\begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ is non-singular only if

$(a-b)^2 + (b-c)^2 + (c-a)^2$

$a^3+b^3+c^3-3abc = \frac{1}{2}(a+b+c)\{(a-b)^2$

$(a^2+b^2+c^2-ab-bc-ca)$

(A) ~~$a \neq b$, $b \neq c$ and $c \neq a$,~~ (B) $a \neq b$ or $b \neq c$ or $c \neq a$ (C) $a+b+c \neq 0$

(D) none of these.

4. $\log \tan 1^0 + \log \tan 2^0 + \ldots + \log \tan 89^0 =$

(A) 0 (B) 1 (C) ∞ (D) none of these

5. Let $x_n = Cos\, \frac{\pi}{2^n} + i\, sin\, \frac{\pi}{2^n}$, then the value of the product

$x_1 . x_2 . x_3 . \ldots \to \infty$ will be $\cos \pi + i \sin \pi$

(A) 0 (B) 1 (C) -1 (D) π.

6. Let $f(x)$ be a function satisfying the condition $f(x) = f(-x)$ for all real x. If $f'(0)$ exists, then $f'(0)$ will be

(A) 0 (B) -1 (C) 1 (D) none of these.

7. In what ratio should a given line be divided into 2 parts so that the area of the rectangle contained by them is maximum?

(A) 1 : 1 (B) 1 : 3 (C) 3 : 2 (D) $(\sqrt{5}-1) : (\sqrt{5}-1)$

8. Given that

$$\int_0^{\infty} \frac{x^2}{(x^2+a^2)(x^2+b^2)(x^2+c^2)}\,dx = \frac{\pi}{2(a+b)(b+c)(c+a)}$$

the value of $\int_0^{\infty} \frac{dx}{(x^2+4)(x^2+9)}$ is

(A) $\frac{\pi}{10}$ (B) $\frac{\pi}{20}$ (C) $\frac{\pi}{[illegible]}$ (D) $\frac{\pi}{60}$.

9. The derivative of $f(x) = |x|^3$ at $x = 0$ is

(A) -1 (B) nonexistent (C) 0 (D) 1/2.

10. Coefficient of x^{20} in the expansion of

$$\left(\frac{1-x^{20}}{(1-x)^2} - \frac{20x^{20}}{(1-x)}\right)$$ is

(A) 0 (B) 1 (C) 20 (D) none of the

11.

$$\lim_{x \to 0} \frac{\tan^3 x}{x^2}$$

(A) is 0 (B) is 1 (C) is ∞ (D) does not exist.

12. A function $f(x)$ is defined as follows :

$$f(x) = \begin{cases} 3 + 2x & for \quad -\frac{3}{2} \le x < 0 \\ 3 - 2x & for \quad 0 \le x < \frac{3}{2} \\ -3 - 2x & for \quad x \ge \frac{3}{2} \end{cases}$$

then $f(x)$ is

(A) continuous at both $x = 0$ and $x = \frac{3}{2}$

(B) continuous at $x = 0$ but discontinuous at $x = \frac{3}{2}$

(C) discontinuous at $x = 0$ but continuous at $x = \frac{3}{2}$

(D) discontinuous at both $x = 0$ and $x = \frac{3}{2}$

13. The solution of the differential euqation

$$(x - y + 5)dy = (5x + 5y - 1)dx$$

is

(A) $x + y + 4\ln(x + y) = c$

(B) $5x + y - 4\ln(x + y) = c$ $-5x + y + 4\log(x+y+1) = c$

(C) $5x - y + 4\ln(x + y) = c$

(D) none of thes

14. The function $f(x) = \frac{\ln(\pi + x)}{\ln(e + x)}$ is

(A) decreasing on $(0, \infty)$

(B) increasing on $(0, \infty)$

(C) decreasing on $(0, \pi/e)$ and increasing on $(\pi/e, \infty)$

(D) increasing on $(0, \pi/e)$ and decreasing on $(\pi/e, \infty)$

15. For the following linear programming problem :

$$\text{maximise} \quad 2x_1 + 3x_2$$

Subject to

$$2x_1 - x_2 \geq 2$$
$$x_1 + x_2 \leq 3$$
$$x_1 - x_2 \leq -1$$
$$x_1, x_2 \geq 0$$

Which one of the following alternatives is true?

(A) the problem has an optimal solution at $(x_1 = 0, x_2 = 3)$

(B) the problem has an optimal solution at $(x_1 = 1, x_2 = 2)$

(C) the problem does not have an optimal solution

(D) none of these

Syllabus and Sample Questions for MEII (Economics) 2000

Syllabus : MEII

Microeconomics : Consumer theory. Producer theory. Market forms. Welfare economics. *Macroeconomics :* National income accounting, simple model of income determination and multiplier: Banking, money supply and inflation. Applicants are expected to have familarity of these topics at the level of Samuelson and Nordhous (Economics), and Lipsey (Positive Economics).

1 Instruction for question numbers 1(a) - (d). For each of the following questions four alternative answers are provided. Choose the answer that you consider to be the most appropriate for a question and write it down in your answer book.

(a) As we move downward along the demand curve, price elasticity of demand will fall if

(A) the demand function is downward sloping

(B) the demand function is downward sloping and concave

(C) the demand function is downward sloping and convex

(D) the demand function is downward sloping and a straight line

(b) Consider the monopoly equilibrium, given the market demand $P = 20 - Q$ and the cost function $C = 15 + 10Q$ (P = price, Q = quantity and C = total cost). Now the government decides to impose a lump sum tax of the amount 15. Then the tax revenue of the government will be

(A) 25; (B) 15; (C) 10; (D) 0

(c) Suppose that on the day before christmas the local garden centre has still a number of cut christmas trees for sale which will be thrown away at zero cost if they are not sold that day. Each tree costs the florist Rs.5/- . What price shoule he/she set to maximize profit?

(A) a price that gives the greatest profit per tree sold:

(B) a price just low enough to make sure that all trees will be sold:

(C) a price that maximizes total receipts;

(D) a price which is at least greater than 5.

(d) Consider the Simple Keynesian Model as given below.

$C = a + bY$

$G = \bar{G}$

$I = \bar{I}$

$X = \bar{X}$

$M = .1Y$

where C stands for consumption demand, G is government expenditure, $\bar{I}$ is autonomous investment, X is export demand and M stands for import.

Which one of the following statements is necessarily false ?

(A) $mpc > 1$

(B) $mpc < 1$

(C) $.5 < mpc < .7$

(D) none of the above

2. (a) Marginal cost (MC) of producing Q units of output is $c(Q) = a + bQ + cQ^2$; $(a, b, c) > 0$. Derive the expression for average variable cost (AVC). Draw the corresponding MC and AVC relations in the same diagram.

(b) A firm has a production function $q = xy$ where x and y are the amounts of two inputs required to produce q units of an output. The per unit price for each factor is 1. If the minimum cost of production is 4, what is q equal to?

3. A monopolist produces in the short run in two separate plants, the total cost functions of which are

$$TC_1 = ax_1^2 + bx_1 + c$$
$$TC_2 = ax_2^2 + dx_2 + e$$

where x_i is the output produced in plant $i (i = 1, 2)$ and a, b, c, d and e are positive constants. The monopolist's profit maximizing equilibrium requires some output from each plant. Show that the plant producing the greater quantity of output will have the lower average variable cost. 8

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4. A firm has two investment projects. In the first project it has to spend Rs.1000 in the current year and it expects to get a return of Rs.100 per year for ever from next year onwards. In the second project it has to spend Rs.2000 in the current year and it expects to get a return of Rs 100 per year for ever from next year onwards.

Draw the demand curve for investment (in rupees) of the firm at all possible market rates of interest.

5. An economy has three classes of people : poor, middle class and rich. A poor person earns Rs.500 per month and has to spend it all on consumption. A middle class person earns Rs.2000 per month of which Rs.1500 is spent and the rest is saved. A rich person earns Rs.10,000 per month and consumes 80% of it, saving the rest.

   (a) Calculate the average propensity to save (aps) in this economy if 20% of the people are poor and 50% are in the middle class.

   (b) Suppose that over time some people move from the poor category to the middle class category and an equal number of people from the middle class category to the rich category. How ~~does this~~ affect aps?

6. Two friends R and S are rational individuals. They enter a food shop for mid-day meal. At the food shop a customer is served, either (1) a fixed meal of 80 gms of rice and 50 gms of dal at a price Rs.15, or (2) any quantity of rice and dal at prices Re.1 and Rs.2 per 10 gms. respectively. R and S place different orders -- viz. R orders a fixed meal whereas S orders 120 gms of rice and 15 gms of dal.

   (a) Using suitable diagrams explain R and S's decisions:

   (b) Suppose the prices of the fixed meal are rice and kept unchanged and the price of dal is raised. Will R and S change their decisions? Explain your answer.

7. A utility maximizing individual lives in a world with only two goods, 1 and 2. His utility function is $U(x_1, x_2) = x_1 x_2$ where x_i is the quantity of good $i, i = 1, 2$. His income is Rs.650 and the prices of goods 1 and 2 are Rs.8 and Rs.2 respectively. The individual is given an opportunity of joining a club whose members can buy good 1 at 50% of the existing price. This is the individual's only benefit from club membership. What is the maximum amount he would just be prepared to pay as membership fees?

8. A farmer finds that for every 100 bushels of corn that he sets aside this year, he can reap 110 bushels next year. He has now 1000 bushels. His utility depends on the quantity of corn he consumes this year and next, according to the function $W = C_0 C_1$, where C_0 is consumption this year and C_1 consumption next year. (Assume corn is perfectly divisible.)

   (a) If there is no capital market, how much should he consume this year and the next to maximum utility?

   (b) If he can borrow any amount at 5% p.a in 1st pd.; how much should he consume in both years?